The Coffee Law: Information and Reasoning Ceilings for Tool-Using Agent Collectives

Manish Bhatt
Affiliation to be completed
email to be completed

***Abstract*—Multi-agent systems built from large language models (LLMs) are often expected to produce knowledge beyond any single agent through debate, critique and voting. We give information-theoretic ceilings on what such collectives can achieve, independent of how the agents work. First, the *Coffee Law*: for any number of agents, any communication protocol and any adaptive tool-query policy, the information the collective holds about an external truth Θ is at most $\min\{H(\Theta),\, I(\Theta; D_0) + \sum_t C_t\}$, where D_0 is the starting data and C_t is the Shannon capacity of round t's tool calls. Fano's inequality turns this into a floor on error and a lower bound on the number of tool rounds. Second, a *reasoning ceiling*: the information that computationally bounded agents can *use* is at most what they *hold*, with the gap equal to an expected KL divergence; computation, including debate, can raise usable information but never available information. Third, *computational bounds* on closing that gap: only a collective's total compute matters, parity requires reading every input, and under standard cryptographic assumptions some fully available information is unusable by any polynomial-time collective. We then extend the law to compute-gated channels (the formal content of sandboxing), worst-case tasks, acting, communication between agents, a drifting truth, adversaries and continuous quantities. The ceilings account for published findings that extra test-time compute does not help on knowledge-intensive tasks, that debate helps on reasoning tasks, and that debate performs about as well as voting at equal compute. Every finite example is verified exactly in public code.**

***Index Terms*—multi-agent systems, large language models, tool use, mutual information, channel capacity, Fano's inequality, usable information**

I. Introduction

LLM-based agent systems combine many model instances that post to a shared board, debate, vote and call tools such as search engines, code interpreters and APIs [1]–[3]. A common expectation is that such a collective can know more than any member. This paper asks how much more, and from where.

We answer with two ceilings that hold for *any* agents—LLMs, people or programs—because the proofs never inspect the agents:

- **The Coffee Law** (Theorem 1). A collective can never know more about the world than its starting data plus what its tools can carry. Communication and internal computation add nothing to this budget; only starting data and informative tool calls raise it. The number of agents does not appear.
- **The reasoning ceiling** (Theorem 2). What computationally bounded agents can *use* is at most what they *hold*. Reasoning, including debate, closes this gap but never passes it.

Together they give the chain

$$\underbrace{I_{\mathcal{V}}(D_T \to \Theta)}_{\text{usable}} \le \underbrace{I(\Theta; D_T)}_{\text{available}} \le \underbrace{\min\{H(\Theta),\, B_T\}}_{\text{budget}}, \qquad (1)$$

with $B_T = I(\Theta; D_0) + \sum_{t<T} C_t$, where reasoning raises the left term and tools raise the right one. Our contributions are: (i) the capacity budget for tool-using collectives with adaptive queries, with a Fano error floor and a tightness example; (ii) the combination of available and usable information into one chain, with an exact expression for the reasoning gap; (iii) computational bounds on closing that gap, from above by total compute and from below unconditionally (parity) and conditionally (a hard-core bit); (iv) extensions that relax each modeling assumption (Section VII); (v) an account of several published empirical findings on debate and test-time compute (Section VIII); and (vi) exact numerical verification of every finite example.[1]

The ingredients—the chain rule, the data-processing inequality (DPI), channel capacity and Fano's inequality—are classical [4], [5]. The contribution is their assembly into a model of tool-using agent collectives and the resulting separation of information-bound from compute-bound tasks.

II. Model

All variables are discrete with finite support; logarithms are base 2. Θ is a fixed external truth. D_t is the collective's *full record* at the start of round t: the initial data D_0 (every agent's context and weights) plus every post, private state, query and tool result since. Each agent's state and the board are functions of D_t. In round t, fresh internal randomness ξ_t (sampling, scheduling) is drawn; the round's tool calls $Q_t = g_t(D_t, \xi_t)$ are issued under any policy g_t; results $O_{t+1} \sim K_t(\cdot \mid \Theta, Q_t, D_t)$ return; communication and computation use fresh randomness ζ_t; and

$$D_{t+1} = (D_t, \xi_t, Q_t, O_{t+1}, \zeta_t).$$

Assumptions. (A1) *No side channels*: Θ reaches the collective only through D_0 and tool results. (A2) *Truth-independent noise*: $\xi_t \perp (\Theta, D_t)$ and $\zeta_t \perp (\Theta, D_t, \xi_t, Q_t, O_{t+1})$. (A3) *Tools are channels from* Θ: given (Θ, Q_t, D_t), O_{t+1} is independent of ξ_t with law K_t; tools may be adaptive or stateful. (A4) *Fixed truth*: actions do not change Θ.

[1] Code, proofs and checks: https://github.com/mbhatt1/coffeelaw.

Capacity. For a query q and history d, $\theta \mapsto K_t(\cdot \mid \theta, q, d)$ is a channel with capacity $\mathrm{Cap}(K_{t,q,d}) = \max_{P_\Theta} I(\Theta; O)$. Let $C_t := \sup_{q,d} \mathrm{Cap}(K_{t,q,d}) \le \log |\mathcal{O}_t|$, where $\mathcal{O}_t$ is the set of possible results in round t.

III. Conservation and Synergy

Proposition 1 (Conservation)**.** *If no tools are called ($C_t = 0$ for all t), then $I(\Theta; D_T) \le I(\Theta; D_0)$.*

This is the special case of Theorem 1 with no tools; it also follows directly from the DPI, since D_T is a function of D_0 and Θ-independent noise. Debate, critique and voting therefore add no information about Θ.

Proposition 2 (Synergy)**.** *Let B_1, B_2 be i.i.d. uniform bits and $\Theta = B_1 \oplus B_2$. Then $I(\Theta; B_1) = I(\Theta; B_2) = 0$ but $I(\Theta; B_1, B_2) = 1$.*

Pooling can make jointly held information accessible that no member can use alone; this is the failure of submodularity of $S \mapsto I(\Theta; S)$ studied in partial information decomposition [6]. The bit was present in D_0 throughout: emergence rearranges the budget without enlarging it.

IV. The Coffee Law

Theorem 1 (Coffee Law)**.** *Under (A1)–(A4), let $B_T := I(\Theta; D_0) + \sum_{t<T} C_t$. Then*

$$I(\Theta; D_T) \le \min\{H(\Theta),\, B_T\},$$

and the same bound holds for every output Y of the collective (answer, board, vote, any agent's state). For any estimator $\hat{\Theta}$ computed from the record, with $|\Theta| \ge 3$,

$$\Pr(\hat{\Theta} \ne \Theta) \;\ge\; \frac{H(\Theta) - \min\{H(\Theta), B_T\} - 1}{\log(|\Theta| - 1)}.$$

Proof. By the chain rule, $I(\Theta; D_{t+1}) - I(\Theta; D_t)$ equals

$$I(\Theta; \xi_t | D_t) + I(\Theta; Q_t | D_t, \xi_t) + I(\Theta; O_{t+1} | S) + I(\Theta; \zeta_t | S, O_{t+1}),$$

with $S = (D_t, \xi_t, Q_t)$. The first and last terms vanish by (A2); the second vanishes because Q_t is a function of (D_t, ξ_t). For the third, fix $S = s = (d, x, q)$. By (A3), given $S = s$ the pair (Θ, O_{t+1}) has $\Theta \sim p(\cdot|s)$ and $O_{t+1} \sim K_t(\cdot|\Theta, q, d)$, so $I(\Theta; O_{t+1} | S = s)$ is the mutual information of channel $K_{t,q,d}$ under input law $p(\cdot|s)$, at most its capacity and hence at most C_t. Averaging over s and summing over rounds gives $I(\Theta; D_T) \le B_T$; also $I(\Theta; D_T) \le H(\Theta)$. Outputs are functions of D_T and independent noise, so the DPI extends the bound to them. For the error floor, Fano's inequality [4] gives $H(\Theta|\hat{\Theta}) \le 1 + \Pr(\hat{\Theta} \ne \Theta) \log(|\Theta| - 1)$, and the DPI gives $H(\Theta|\hat{\Theta}) \ge H(\Theta) - \min\{H(\Theta), B_T\}$. □

Adaptivity is already accounted for: the query policy only selects the input law $p(\cdot|s)$ and the channel $K_{t,q,d}$, and capacity is the maximum over both. This mirrors the classical fact that feedback does not increase the capacity of a memoryless channel [7], [8] and the use of Fano's inequality for active-learning lower bounds [9].

TABLE I
Exact information about Θ after two tool rounds (budget $2C$).

Policy	$I(\Theta; D_2)$ (bits)	Budget $\min\{H, 2C\}$
Query bit 0, then bit 1	1.0620	1.0620
Adaptive, may repeat a bit	0.9820	1.0620
Either policy, plus a debate step	unchanged	—
Both bits, noise-free tool	2.0000	2.0000

Corollary 1 (Rounds needed)**.** *If Θ is uniform on 2^k values ($k \ge 2$), $I(\Theta; D_0) = 0$ and $C_t \le C$ for all t, then reaching $\Pr(\hat{\Theta} \ne \Theta) \le \varepsilon$ requires $T \ge ((1 - \varepsilon)k - 1)/C$ tool rounds, regardless of the number of agents or how they communicate.*

Corollary 2 (Agents do not appear)**.** *Synergy, debate, voting and the number of agents N are absent from B_T. More agents raise the budget only by issuing more or better tool calls, i.e. by raising C_t.*

Tightness. Let Θ be uniform on $\{0, 1, 2, 3\}$ and let a call $q \in \{0, 1\}$ return bit q of Θ through a binary symmetric channel with flip probability 0.1, so $C = 1 - h(0.1) \approx 0.531$ bits. Table I gives exact values. Querying both bits attains the budget; an adaptive policy that sometimes re-queries the same bit falls below it; a debate step adds nothing.

Exact accounting and when the law is tight. The proof gives the identity $I(\Theta; D_T) = I(\Theta; D_0) + \sum_t \mathbb{E}_s[I(\Theta; O_{t+1} | S_t = s)]$, and each term is at most the capacity of the channel actually used, which is at most $\min\{\log |\mathcal{O}_t|, \log |\mathrm{supp}\, \Theta|\}$. Equality in Theorem 1 holds iff almost every posterior is a capacity-achieving input for the chosen channel and that channel attains C_t. For rich tools the binding quantity is the realized gain; the capacity form bites for low-capacity tools. *Example:* a test run returns pass or fail, so it carries at most 1 bit. Locating one bad commit among n with error at most ε needs at least $(1 - \varepsilon) \log n - 1$ runs (Corollary 1), whatever agents run the search; binary search (`git bisect`) uses $\lceil \log n \rceil$. A flaky test that misreports with probability p raises the minimum by the factor $1/(1 - h(p))$.

V. The Reasoning Ceiling

Holding information is not the same as being able to use it: the prime factors of N are fully determined by N, yet finding them may take enormous computation. Following Xu et al. [10], we model bounded agents by a predictive family.

Definition 1. A *predictive family* $\mathcal{V}$ is a set of predictors f mapping a record d to a distribution $f[d]$ on Θ, containing the constant prior predictor $f[d] = p_\Theta$. Define $H_\mathcal{V}(\Theta|D) := \inf_{f \in \mathcal{V}} \mathbb{E}[-\log f[D](\Theta)]$ and the *usable information* $I_\mathcal{V}(D \to \Theta) := H(\Theta) - H_\mathcal{V}(\Theta|D)$.

Xu et al. use the family's own no-input entropy as the baseline; with the prior predictor in $\mathcal{V}$, as assumed here, the baseline is $H(\Theta)$.

Theorem 2 (Reasoning ceiling)**.** (a) $0 \le I_{\mathcal{V}}(D \to \Theta) \le I(\Theta; D)$*, and the* reasoning gap *is*

$$I(\Theta; D) - I_{\mathcal{V}}(D \to \Theta) = \inf_{f \in \mathcal{V}} \mathbb{E}_D\big[D_{\mathrm{KL}}\big(p(\cdot|D) \,\|\, f[D]\big)\big].$$

(b) *Under (A1)–(A4), the chain* (1) *holds for the record* D_T.

(c) *There exist* $\mathcal{V}$ *and a deterministic* g *with* $I(\Theta; g(D)) = I(\Theta; D)$ *but* $I_{\mathcal{V}}(g(D) \to \Theta) > I_{\mathcal{V}}(D \to \Theta)$*. For every* g*,* $I_{\mathcal{V}}(g(D) \to \Theta) \le I(\Theta; D)$.

Proof. (a) The prior predictor gives $H_{\mathcal{V}}(\Theta|D) \le H(\Theta)$. For any f, the cross-entropy identity gives $\mathbb{E}[-\log f[D](\Theta)] = H(\Theta|D) + \mathbb{E}_D D_{\mathrm{KL}}(p(\cdot|D) \| f[D])$; take the infimum and subtract from $H(\Theta)$. (b) Part (a) and Theorem 1. (c) The last claim is (a) followed by the DPI. For the strict example take Proposition 2 with $D = (B_1, B_2)$ and the "one-look" family $\mathcal{V}_1$ of predictors that read a single coordinate of the record. Each coordinate of D is independent of Θ, so $I_{\mathcal{V}_1}(D \to \Theta) = 0$. After one reasoning step $g(D) = (B_1 \oplus B_2, B_2)$ the first coordinate equals Θ, so $I_{\mathcal{V}_1}(g(D) \to \Theta) = 1 = I(\Theta; g(D)) = I(\Theta; D)$. □

Part (c) is the observation of [10] that usable information violates the DPI, placed in the collective setting.

Corollary 3 (Debate adds reasoning, not facts)**.** *In a closed collective every post is a function of* D_0 *and noise, so it adds no available information (Proposition 1). It can add usable information: an agent that posts an intermediate result hands others a computation outside their family. The gain is capped by* $I(\Theta; D_0)$.

Two regimes. A task is *information-bound* when the right inequality of (1) binds (forecasting, facts about the world, diagnosing a live system): only tools and better starting data help. It is *compute-bound* when the left gap is large (mathematics, proofs, code, planning from known facts): more reasoning helps. A code interpreter applied to data already held enlarges $\mathcal{V}$; a tool that observes the world raises C_t. In the noisy example of Table I, the record holds 1.0620 bits but a one-look agent can use only 0.2655 of them; the remaining 0.7965 bits are a reasoning gap that more observations cannot close.

Imperfect agents. Real agents fall further below the chain; the following are proved in the accompanying notes (footnote 1). For a decision with utility range U, acting on a belief $\hat{b}$ instead of the correct posterior b^* costs at most $2U\,\mathrm{TV}(b^*, \hat{b}) + \delta \le 2U\sqrt{(\ln 2/2)\, D_{\mathrm{KL}}(b^* \| \hat{b})} + \delta$, where δ is the agent's suboptimality given its own belief. A summary M of data D loses exactly $\mathbb{E}\, D_{\mathrm{KL}}(p(\cdot|D) \| p(\cdot|M)) = I(\Theta; D) - I(\Theta; M)$. And a message with zero information about Θ can still flip a decision: an agent with belief $(0.6, 0.4)$ in favor of the truth that gives weight α to a constant false claim answers wrongly once $\alpha > 1/6$.

VI. Computational Bounds on Reasoning

How much computation does closing the reasoning gap take? Fix a machine model (a word RAM) and let $\mathcal{V}_S$ be the predictors whose probabilities can be computed, and which can be sampled from, in at most S steps, reading one symbol of the record per step. Each $\mathcal{V}_S$ contains the prior predictor and $S \mapsto I_{\mathcal{V}_S}$ is nondecreasing and at most $I(\Theta; D)$. A collective's *total compute* is the sum over agents and rounds of their steps, with each board read or write counted as a step.

Theorem 3 (Compute ceiling)**.** *A predictor output by a collective with total compute* S *lies in* $\mathcal{V}_{\kappa S}$ *for a constant* κ *depending only on the machine model, so the collective's usable information is at most* $I_{\mathcal{V}_{\kappa S}}(D \to \Theta) \le I(\Theta; D)$.

Proof. One machine can run the agents step by step in a fixed interleaving, keeping each agent's memory and the board in separate regions, at $O(1)$ cost per simulated step; it outputs the same predictor in κS steps. □

The number of agents enters only through S: parallelism buys wall-clock time, not capability.

Lemma 1. *Sampling* $\hat{\Theta} \sim f[D]$ *succeeds with probability at least* $2^{-\mathbb{E}[-\log f[D](\Theta)]}$*. If* $H(\Theta) \le 1$ *and* f *achieves usable information* δ*, it succeeds with probability at least* $2^{\delta - 1} \ge \frac{1}{2} + \frac{\ln 2}{2}\delta$.

Proof. The success probability is $\mathbb{E}[f[D](\Theta)]$; apply Jensen's inequality to $-\log$, and use $2^{\delta-1} = \frac{1}{2}e^{\delta \ln 2}$. □

Theorem 4 (Compute lower bounds)**.** *(a) Let* $D = (B_1, \ldots, B_n)$ *be uniform bits and* $\Theta = \bigoplus_i B_i$*. Any predictor, or any collective counting all agents' reads, that reads at most* $n - 1$ *bits of* D*, even adaptively, has zero usable information; reading all* n *gives one bit. (b) Assume one-way permutations exist. For a one-way permutation* f*, uniform* $x, r \in \{0,1\}^n$*,* $D = (f(x), r)$ *and* $\Theta = \langle x, r \rangle \bmod 2$*, we have* $I(\Theta; D) = H(\Theta)$*, yet every polynomial-time family has* $I_{\mathcal{V}_{\mathrm{poly}}}(D \to \Theta) \le \log(1 + 2\varepsilon(n))$ *with* ε *negligible.*

Proof. (a) Conditioned on any transcript of fewer than n reads, the unread bits are uniform and independent, so Θ is independent of the transcript; the output is a function of the transcript and independent randomness, so Theorem 2(a) gives zero. Posts are functions of their authors' reads, so the bound covers the union of a collective's reads. (b) Θ is a function of D because f is a bijection. By the Goldreich–Levin theorem [11], every polynomial-time algorithm guesses Θ with probability at most $\frac{1}{2} + \varepsilon(n)$; by Lemma 1, every polynomial-time predictor has cross-entropy at least $1 - \log(1 + 2\varepsilon)$, and $H(\Theta) \le 1$. □

By Theorem 3, part (b) holds for any polynomial-time collective, however many agents it has. Part (a) shows splitting work can only reduce wall-clock time: N agents can each read n/N bits and post block parities, but the total is still at least n reads. Between these extremes no general efficient bound is

possible without resolving open questions in complexity theory. The parity bounds and Lemma 1 are checked numerically in the code; part (b) is conditional.

VII. Extensions

Each result below relaxes one modeling assumption. Full proofs and exact checks are in the accompanying notes (footnote 1).

Theorem 5 (Compute-gated channels)**.** *Let $A_S(d)$ be the queries a collective can issue from record d within total compute S. Then $I(\Theta; D_T) \le I(\Theta; D_0) + \sum_t \sup_{d,\, q \in A_S(d)} \mathrm{Cap}(K_{t,q,d})$, which is nondecreasing in S. If a tool reveals a uniform 2-bit Θ only after a uniform k-bit password is found, a collective trying S passwords has exactly $I(\Theta; D) = \min\{1, S/2^k\} \cdot 2$.*

Hence "reasoning never raises available information" holds only for a fixed set of reachable channels: reasoning that discovers channels (endpoints, credentials, exploits) raises the budget. A *sandbox* is a guarantee that $A_S(d) \subseteq A_{\text{allowed}}$ for every feasible S; an escape is a feasible S violating it. Recently reported incidents in which models under evaluation left their test sandbox to retrieve benchmark answers from an outside server are instances: the answers were not inside, so the only route was an unallowed channel.

Theorem 6 (Worst case)**.** *For $M \ge 3$ possible truths with no prior, initial data $D_0 \sim K_0(\cdot|\theta)$ of capacity C_0, and any strategy, $\max_i \Pr_{\theta_i}(\hat{\Theta} \ne \theta_i) \ge (\log M - \min\{\log M, C_0 + \sum_t C_t\} - 1)/\log(M-1)$.*

This is Fano's method under the uniform prior: for any family of tasks the collective cannot separate within its budget, there is a specific task on which it errs this often.

Theorem 7 (Acting is bounded by knowing)**.** *In a fixed unknown environment W with $M \ge 3$ actions, a unique best action $A^* = a^*(W)$ and gap Δ, actions that may themselves be tool calls satisfy $\mathbb{E}[\mathrm{regret}_t] \ge \Delta\,(H(A^*) - \min\{H(A^*), I(W; D_0) + \sum_{s<t} C_s\} - 1)/\log(M-1)$.*

The action is an estimator of A^*, and $I(A^*; D_t) \le I(W; D_t)$. Taking W to be a user's hidden specification covers open-ended tasks: an agent cannot reliably build what it has not learned the user wants.

Theorem 8 (Communication ceiling)**.** *(a) If the messages reaching agent 1 take at most 2^L values, then $I(\Theta; D^1, \Pi) \le I(\Theta; D^1) + L$. (b) If agents 1 and 2 hold uniform $x, y \in \{0,1\}^n$ and $\Theta = \langle x, y \rangle \bmod 2$, then $I(\Theta; x, y) = H(\Theta)$, yet any protocol exchanging L bits gives $I(\Theta; x, \Pi) \le \log(1 + 2^{L+1-n/2})$, even with unlimited computation.*

Part (b) combines the discrepancy bound for inner product (Lindsey's lemma) [12], [13] with Lemma 1. The collective holds the answer, but no member learns it without about $n/2$ bits of communication: a third ceiling, distinct from information and compute.

Theorem 9 (Drifting truth)**.** *If Θ_t is a Markov chain unaffected by the agents, with strong data-processing constant $\eta < 1$ [14], and tools observe the current state, then $x_t = I(\Theta_t; D_t)$ satisfies $x_{t+1} \le \eta(x_t + C_t)$, so $x_t \le \eta^t x_0 + \eta C/(1-\eta)$ when $C_t \le C$.*

In a changing world knowledge saturates however long agents run. For a bit flipping with probability ρ, $\eta = (1-2\rho)^2$.

Theorem 10 (Provenance)**.** *If every post is stored with its author, adversarial posts M cannot reduce available information: $I(\Theta; D, M) \ge I(\Theta; D)$. Without provenance, one adversary can erase it: if an honest agent posts a fair bit Θ and an adversary posts $1 - \Theta$, the unattributed record carries 0 bits.*

Adversaries can hide information but not destroy it, as long as messages are attributed.

Theorem 11 (Continuous truth)**.** *For real Θ with differential entropy $h(\Theta)$, the budget of Theorem 1 holds with capacities under input constraints (e.g. $\frac{1}{2}\log(1 + P/\sigma^2)$ for a Gaussian tool), and every estimator has $\mathbb{E}[(\Theta - \hat{\Theta})^2] \ge (2\pi e)^{-1}\, 2^{2(h(\Theta) - I(\Theta; D_0) - \sum_t C_t)}$.*

This is the estimation counterpart of Fano's inequality [4]; it holds with equality for a Gaussian truth observed in Gaussian noise.

VIII. Consistency with Empirical Findings

The chain (1) accounts for several published results.

Information-bound tasks resist reasoning. Across 14 reasoning models on closed-book knowledge benchmarks, more test-time compute does not reliably raise accuracy and often increases hallucination [15]; the authors argue that compute-only scaling of a fixed model cannot add information about the answer, which is Proposition 1. Retrieval and tools, which raise C_t, are the standard remedy for knowledge-intensive tasks [2], [3], [16]. Intrinsic self-correction without external feedback likewise often fails to improve reasoning [17].

Compute-bound tasks benefit from debate. Multi-agent debate improves mathematical and strategic reasoning [1], consistent with debate closing a reasoning gap.

Debate is worth its reasoning gain. Debate does not reliably outperform self-consistency or ensembling and costs more [18], and most of its gains are attributable to majority voting [19]. If debate's value is computation, equal computation spent another way should do as well.

Two apparent tensions. Debate is also reported to improve *factuality* [1]. The facts were in the weights, i.e. in D_0, so debate made stored information usable without adding any. Debate is proved not to improve expected correctness for Bayesian agents [19]; such agents have an unrestricted $\mathcal{V}$, so their reasoning gap is zero and Theorem 2 predicts no gain.

Testable predictions. (P1) Error rate versus the number of informative tool calls stays above the floor of Theorem 1. (P2) At matched compute, debate improves compute-bound tasks but not information-bound ones. (P3) Redundant or low-capacity tool calls add little, even when numerous.

IX. Related Work

The closed-system case for multi-agent debate is proved via the DPI in [20], and debate is modeled as a belief martingale in [19]. Yang et al. [21] bound multi-agent performance by task uncertainty and a sum of per-agent channel informations, the same "$\min\{H, \sum \text{channels}\}$" shape with agents rather than tool calls as the channels. Usable information under computational constraints is due to [10]; synergy is studied in partial information decomposition [6]; directed information and feedback in [8]; and Fano-based lower bounds for active learning in [9]. To our knowledge, the per-call capacity budget for tool-using collectives with adaptive queries, and its combination with usable information, have not been stated before.

X. Limitations

Section VII relaxes each assumption, but some gaps remain. Outcomes the agents themselves change are not covered. Theorem 1's equality condition is characterized, but no general algorithm achieves it. The computational bounds are task-specific: Theorem 4 gives one unconditional and one conditional case, not a general formula for the compute a given gap requires. Assumption (A1) requires accounting for everything a model absorbed in training, and $I(\Theta; D_0)$ and C_t are hard to estimate for real models and tools; the bounds' value is often structural rather than numerical. We report no new experiments; the predictions of Section VIII are untested here.

XI. Conclusion

A collective of agents can be clever, but it cannot know more than it has seen. The Coffee Law bounds what a tool-using collective can learn by its starting data plus the capacity of its tool calls; the reasoning ceiling bounds what it can use by what it holds; and reaching that ceiling costs computation that only total compute can buy, from linear in the input to beyond any efficient collective, depending on the task. Practically: if the answer is in the data, add reasoning; if it is not, add better observations of the world. Neither substitutes for the other.

References

[1] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 235, 2024.

[2] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *Int. Conf. Learning Representations (ICLR)*, 2023.

[3] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language models can teach themselves to use tools," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[4] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[5] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, pp. 379–423, 623–656, 1948.

[6] P. L. Williams and R. D. Beer, "Nonnegative decomposition of multivariate information," arXiv:1004.2515, 2010.

[7] C. E. Shannon, "The zero error capacity of a noisy channel," *IRE Transactions on Information Theory*, vol. 2, no. 3, pp. 8–19, 1956.

[8] J. L. Massey, "Causality, feedback and directed information," in *Proc. Int. Symp. Information Theory and its Applications (ISITA)*, 1990.

[9] M. Raginsky and A. Rakhlin, "Lower bounds for passive and active learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2011.

[10] Y. Xu, S. Zhao, J. Song, R. Stewart, and S. Ermon, "A theory of usable information under computational constraints," in *Int. Conf. Learning Representations (ICLR)*, 2020.

[11] O. Goldreich and L. A. Levin, "A hard-core predicate for all one-way functions," in *Proc. ACM Symp. Theory of Computing (STOC)*, 1989.

[12] B. Chor and O. Goldreich, "Unbiased bits from sources of weak randomness and probabilistic communication complexity," *SIAM Journal on Computing*, vol. 17, no. 2, pp. 230–261, 1988.

[13] E. Kushilevitz and N. Nisan, *Communication Complexity*. Cambridge University Press, 1997.

[14] R. Ahlswede and P. Gács, "Spreading of sets in product spaces and hypercontraction of the Markov operator," *Annals of Probability*, vol. 4, no. 6, pp. 925–939, 1976.

[15] J. X. Zhao, B. Hooi, and S.-K. Ng, "Test-time scaling in reasoning models is not effective for knowledge-intensive tasks yet," in *Int. Conf. Learning Representations (ICLR)*, 2026, arXiv:2509.06861.

[16] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[17] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in *Int. Conf. Learning Representations (ICLR)*, 2024.

[18] A. Smit, P. Duckworth, N. Grinsztajn, T. D. Barrett, and A. Pretorius, "Should we be going MAD? A look at multi-agent debate strategies for LLMs," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 235, 2024.

[19] H. K. Choi, X. Zhu, and S. Li, "Debate or vote: Which yields better decisions in multi-agent large language models?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[20] K. S. Shin, "The reasoning trap: An information-theoretic bound on closed-system multi-step LLM reasoning," arXiv:2605.01704, 2026.

[21] Y. Yang, C. Qu, M. Wen, L. Shi, Y. Wen, W. Zhang, A. Wierman, and S. Gu, "Understanding agent scaling in LLM-based multi-agent systems via diversity," arXiv:2602.03794, 2026.